

ME563 HW 4

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Problem 1

Problem statement. Incompressible fluid flows past an impermeable flat plate, with a uniform inlet profile $u = U_0$ and a cubic polynomial exit profile $u = U_0 \frac{3\eta - \eta^3}{2}$ where $\eta = \frac{y}{\delta}$.

The CV is the rectangle $0 \leq y \leq \delta$ between the inlet and exit, width b into the page. The flow is steady and incompressible, the bottom face is the impermeable plate, and $u = U_0$ at $y = \delta$. Reynolds transport theorem for a fixed CV, with B the extensive property and $\beta = dB/dm$:

$$\frac{dB_{sys}}{dt} = \frac{d}{dt} \int_{CV} \rho \beta dV + \int_{CS} \rho \beta (\mathbf{u} \cdot \hat{\mathbf{n}}) dA$$

(a)

Problem statement. Determine the volume flow Q across the top surface of the control volume.

Mass ($\beta = 1$, $dm_{sys}/dt = 0$), steady with constant ρ :

$$\int_{CS} \mathbf{u} \cdot \hat{\mathbf{n}} dA = 0 \Rightarrow - \int_0^\delta U_0 b dy + Q + \int_0^\delta u(y) b dy = 0$$

Exit flux, with $dy = \delta d\eta$:

$$\int_0^\delta u b dy = U_0 b \delta \int_0^1 \frac{3\eta - \eta^3}{2} d\eta = U_0 b \delta \cdot \frac{1}{2} \left(\frac{3}{2} - \frac{1}{4} \right) = \frac{5}{8} U_0 b \delta$$

$$-U_0 b \delta + Q + \frac{5}{8} U_0 b \delta = 0 \Rightarrow \boxed{Q = \frac{3}{8} U_0 b \delta \quad (\text{out of the CV})}$$

The plate slows the fluid near the wall, so 3/8 of the inflow is displaced out through the top.

(b)

Problem statement. Determine the drag force exerted on the flow by the solid plate. Assume pressure is uniform, and hence there's no net pressure force.

x -momentum ($\beta = u$), steady. The only x -force on the fluid is the plate force F_x :

$$F_x = \int_{CS} u \rho (\mathbf{u} \cdot \hat{\mathbf{n}}) dA$$

Fluid leaving through the top carries x -momentum ρU_0 per unit volume:

$$\text{inlet: } - \int_0^\delta \rho U_0^2 b dy = -\rho U_0^2 b \delta$$

$$\text{top: } \rho U_0 Q = \frac{3}{8} \rho U_0^2 b \delta$$

$$\begin{aligned} \text{exit: } \int_0^\delta \rho u^2 b dy &= \rho U_0^2 b \delta \int_0^1 \left(\frac{3\eta - \eta^3}{2} \right)^2 d\eta = \frac{\rho U_0^2 b \delta}{4} \int_0^1 (9\eta^2 - 6\eta^4 + \eta^6) d\eta \\ &= \frac{\rho U_0^2 b \delta}{4} \left(3 - \frac{6}{5} + \frac{1}{7} \right) = \frac{17}{35} \rho U_0^2 b \delta \end{aligned}$$

$$F_x = \rho U_0^2 b \delta \left(-1 + \frac{3}{8} + \frac{17}{35} \right) = \rho U_0^2 b \delta \frac{-280 + 105 + 136}{280}$$

$$F_x = -\frac{39}{280} \rho U_0^2 b \delta \approx -0.139 \rho U_0^2 b \delta$$

The plate pulls the flow upstream ($-x$) with magnitude $\frac{39}{280} \rho U_0^2 b \delta$. By Newton's third law, the drag on the plate is equal and opposite ($+x$).

Problem 2

Problem statement. Derive a formula for the horizontal force F required to hold the gate. At sections 1 and 2, the flow is uniform. Neglecting bottom friction. Assume steady incompressible flow with no variation across the width b . Express your final formula in terms of the inlet velocity V_1 , eliminating V_2 .

The CV runs from section 1 to section 2 and contains the gate. The streamlines at 1 and 2 are straight and parallel, so the pressure there is hydrostatic. Atmospheric pressure acts on the whole CS and cancels, so I use gage pressure. F is the force of the gate on the fluid, acting in $-x$ as drawn.

Mass:

$$\rho V_1 h_1 b = \rho V_2 h_2 b \Rightarrow V_2 = V_1 \frac{h_1}{h_2}$$

Hydrostatic pressure forces:

$$F_{p1} = \int_0^{h_1} \rho g (h_1 - y) b dy = \frac{1}{2} \rho g h_1^2 b (+x), \quad F_{p2} = \frac{1}{2} \rho g h_2^2 b (-x)$$

x -momentum, with $\dot{m} = \rho V_1 h_1 b$:

$$\sum F_x = \int_{CS} u \rho (\mathbf{u} \cdot \mathbf{n}) dA \Rightarrow \frac{1}{2} \rho g b h_1^2 - \frac{1}{2} \rho g b h_2^2 - F = \rho V_1 h_1 b (V_2 - V_1)$$

Substituting $V_2 - V_1 = V_1 (h_1/h_2 - 1)$:

$$F = \frac{1}{2} \rho g b (h_1^2 - h_2^2) - \rho b h_1 V_1^2 \left(\frac{h_1}{h_2} - 1 \right)$$

As $V_1 \rightarrow 0$ this reduces to the net hydrostatic force. Because the flow accelerates through the gate ($V_2 > V_1$), the momentum term lowers the force needed to hold the gate.

Problem 3

Problem statement. When a uniform stream flows past an immersed thick cylinder, a broad low-velocity wake is created downstream, idealized as a “V”. Pressures at 1 and 2 are approximately equal. If the flow is two-dimensional and incompressible, with width b into the paper, determine the drag force F exerted on the flow by the cylinder.

The CV is a rectangle from section 1 to section 2, with its top and bottom at $y = \pm H$ in the free stream ($H > L$). Since $p_1 \approx p_2$ and the pressure along the top and bottom is uniform, there is no net pressure force, and shear on the top and bottom is negligible. The only x -force on the fluid is F_x , the force of the cylinder on the flow. By symmetry, I work with the upper half of the wake, $0 \leq y \leq L$:

$$u(y) = \frac{U}{2} \left(1 + \frac{y}{L}\right) = \frac{U}{2}(1 + \eta), \quad \eta = \frac{y}{L}$$

Mass. The wake has a velocity deficit, so a flow Q leaves through the top and bottom:

$$\begin{aligned} Q &= \underbrace{2HU}_{\text{inlet}} - \underbrace{2(H-L)U}_{\text{exit, outside wake}} - \underbrace{2 \int_0^L \frac{U}{2} \left(1 + \frac{y}{L}\right) dy}_{\text{exit, wake}} \quad (\text{per unit } b) \\ &= 2UL - UL \left(1 + \frac{1}{2}\right) \Rightarrow Q = \frac{1}{2} UL b \end{aligned}$$

That flow leaves in the free stream, so it carries x -momentum ρU per unit volume.

x -momentum:

$$F_x = \int_{CS} u \rho (\mathbf{u} \cdot \hat{\mathbf{n}}) dA$$

Wake contribution:

$$2 \int_0^L \rho u^2 dy = 2\rho \frac{U^2}{4} L \int_0^1 (1 + \eta)^2 d\eta = \frac{\rho U^2 L}{2} \left[\frac{(1 + \eta)^3}{3} \right]_0^1 = \frac{7}{6} \rho U^2 L$$

Summing over the CS (per unit b):

$$\frac{F_x}{\rho U^2 b} = \underbrace{-2H}_{\text{inlet}} + \underbrace{\frac{1}{2}L}_{\text{top+bottom}} + \underbrace{2(H-L)}_{\text{exit, outside}} + \underbrace{\frac{7}{6}L}_{\text{exit, wake}} = L \left(\frac{1}{2} - 2 + \frac{7}{6} \right) = -\frac{L}{3}$$

$$F_x = -\frac{1}{3} \rho U^2 b L$$

H drops out, as it should. The cylinder exerts a force of $\frac{1}{3}\rho U^2 b L$ on the flow, directed upstream. The drag on the cylinder is equal and opposite ($+x$).

Problem 4

Problem statement. Consider water waves in a channel of depth H . Suppose that the span-wise width of the channel is large and the waves are two-dimensional. The undisturbed water level $y = 0$, is disturbed by the wave to $y = h(x, t) = A \cos(kx - \omega t)$, where A denotes the amplitude

of the waves. The bottom of the channel is located at $y = -H$, and is rigid. Suppose the velocity field due to the wave is given by:

$$\begin{aligned} u(x, y, t) &= -A\omega \sin(kx - \omega t) \cosh(k(y + H)) / \sinh(kH) \\ v(x, y, t) &= A\omega \cos(kx - \omega t) \sinh(k(y + H)) / \sinh(kH) \end{aligned}$$

Let $\theta = kx - \omega t$ and $C = A\omega / \sinh(kH)$.

(1)

Problem statement. Is this an incompressible flow?

$$\begin{aligned} \frac{\partial u}{\partial x} &= -Ck \cos \theta \cosh k(y + H), & \frac{\partial v}{\partial y} &= Ck \cos \theta \cosh k(y + H) \\ \nabla \cdot \mathbf{u} &= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \boxed{\text{Yes, the flow is incompressible.}} \end{aligned}$$

(2)

Problem statement. Compute the vorticity field associated with this flow.

For planar flow, only ω_z is nonzero:

$$\begin{aligned} \frac{\partial v}{\partial x} &= -Ck \sin \theta \sinh k(y + H), & \frac{\partial u}{\partial y} &= -Ck \sin \theta \sinh k(y + H) \\ \omega_z &= \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \Rightarrow \boxed{\omega = 0} \end{aligned}$$

The flow is irrotational. It is the gradient of the potential $\phi = -\frac{A\omega}{k} \sin \theta \frac{\cosh k(y + H)}{\sinh(kH)}$.

(3)

Problem statement. Find the pathlines for this flow. For this purpose take $A/H = 10^{-3}$ and $kH = 10$ and integrate the equation numerically and show the result as a graph of the computed pathlines. You can show the pathline originating at $x_0 = 0.5$, and $y_0 = 0.0$ at $t = 0$. Choose suitable non-dimensional variables in your calculation and plot, please define the choices you make.

Pathlines satisfy $dx/dt = \mathbf{u}(\mathbf{x}, t)$. I nondimensionalize length by the depth and time by the wave frequency:

$$x^* = \frac{x}{H}, \quad y^* = \frac{y}{H}, \quad t^* = \omega t$$

Since $dx/dt = \omega H dx^*/dt^*$, this gives

$$\begin{aligned} \frac{dx^*}{dt^*} &= -\frac{A}{H} \sin(kH x^* - t^*) \frac{\cosh(kH(y^* + 1))}{\sinh(kH)} \\ \frac{dy^*}{dt^*} &= \frac{A}{H} \cos(kH x^* - t^*) \frac{\sinh(kH(y^* + 1))}{\sinh(kH)} \end{aligned}$$

The equations depend only on $A/H = 10^{-3}$ and $kH = 10$, so k and ω are not needed. I take the initial condition $(x_0^*, y_0^*) = (0.5, 0)$ at $t^* = 0$. The system was integrated over 10 wave periods

($0 \leq t^* \leq 20\pi$) with SciPy's `solve_ivp` (DOP853, $\text{rtol} = 10^{-11}$) in `hw4_problem4.py`. Figure 1(a) shows the required pathline. Figure 1(b) adds pathlines starting at three depths below the surface, plotted as displacement scaled by A .

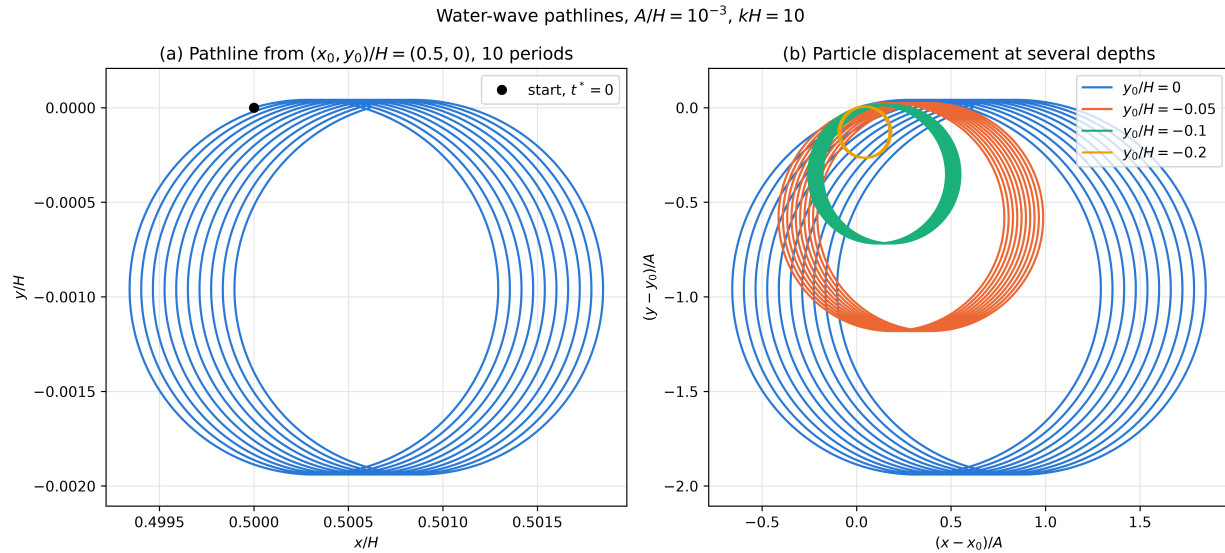


Figure 1: Pathlines for $A/H = 10^{-3}$, $kH = 10$ over 10 wave periods. (a) Pathline from $(x_0, y_0)/H = (0.5, 0)$. (b) Displacement from the starting point, scaled by A , for $y_0/H = 0, -0.05, -0.1, -0.2$.

Observations:

- With $kH = 10$, $\tanh(kH) \approx 1$, so this is a deep-water wave. The particles move on near-circular orbits of radius $\approx A e^{ky_0}$. The radius is about A at the surface and decays quickly with depth, to about $0.14A$ at $y_0/H = -0.2$.
- The orbits do not quite close. Each period, the particle drifts forward slightly, which is Stokes drift, a second-order effect. The drift per period from the integration agrees with the Stokes result $\bar{u}_s = A^2 \omega k \cosh(2k(y + H)) / (2 \sinh^2 kH)$ to within about 3%:

y_0/H	Drift per period / A (numerical)	Drift per period / A (Stokes)
0	0.0611	0.0628
-0.05	0.0227	0.0231
-0.10	0.0084	0.0085
-0.20	0.0011	0.0012

The flow is unsteady, so these pathlines are not the instantaneous streamlines.